

Staging Areas for Data Warehouses

Data warehouse staging areas

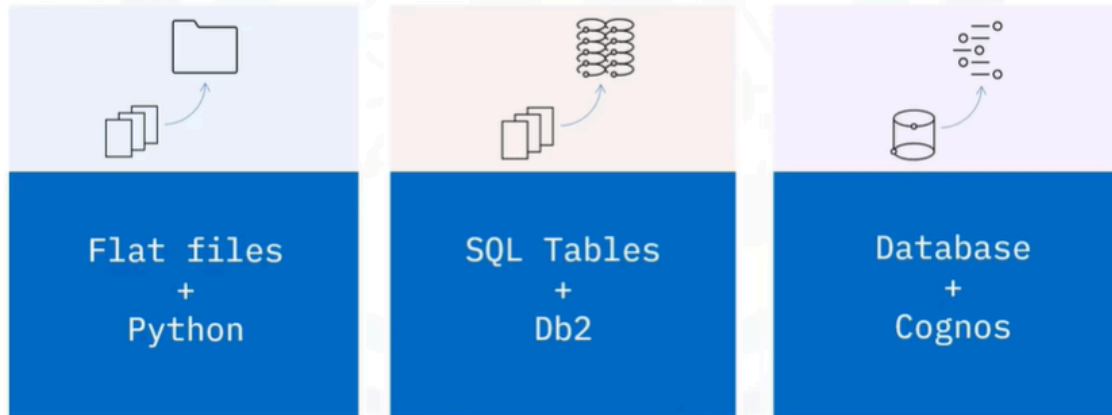
A staging area is:

- Intermediate storage for ETL processing
- Bridge between data sources and the target system
- Sometimes transient
- Sometimes held for archiving or troubleshooting
- Used to optimize and monitor ETL jobs

- What is a data warehouse staging area? You can think of a staging area as an intermediate storage area that is used for ETL processing. Thus, staging areas act as a bridge between data sources and the target data warehouses, data marts, or other data repos. They are often transient, meaning that they are erased after successfully running ETL workflows. However, many architectures hold data for archival or troubleshooting purposes. They are also useful for monitoring and optimizing your ETL workflows.

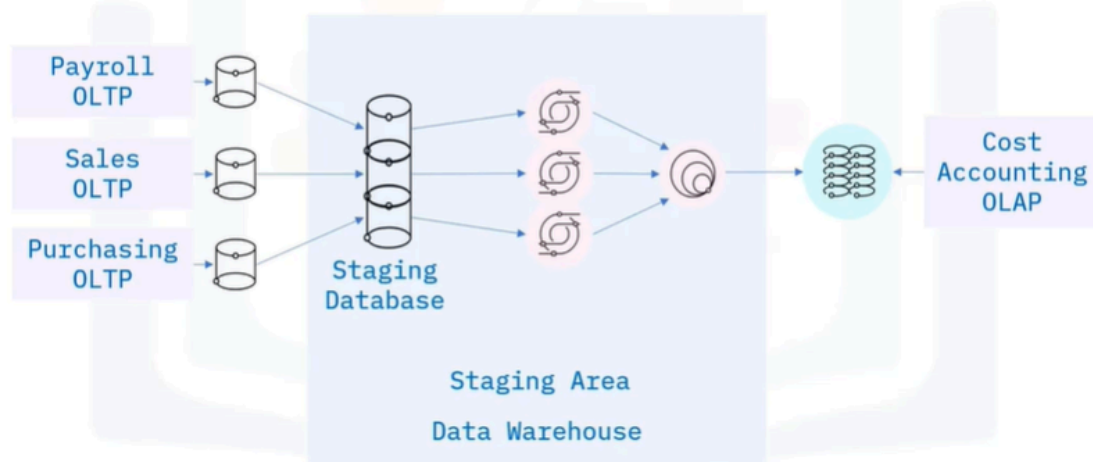
Data warehouse staging areas

Staging areas can be implemented as:



- Staging areas can be implemented in many ways, including: simple flat files, such as csv files, stored in a directory, and managed using tools such as Bash or Python, or a set of SQL tables in a relational database such as Db2, or a self-contained database instance within a data warehousing or business intelligence platform such as Cognos Analytics.

DW staging area example



- Let's explore an example use case, to illustrate a possible architecture for a Data Warehouse containing a Staging Area, which in turn includes an associated Staging Database. Imagine the enterprise would like to create a dedicated "Cost Accounting" Online Analytical Processing system.
- The required data is managed in separate Online Transaction Processing Systems within the enterprise, from the Payroll, Sales, and Purchasing departments. From these siloed systems, the data is extracted to individual Staging Tables, which are created in the Staging Database. Data from these tables is then transformed in the Staging Area using SQL to conform it to the requirements of the Cost Accounting system.
- The conformed tables can now be integrated, or joined, into a single table. The final phase is the loading phase, where the data is loaded into the target cost-accounting system.

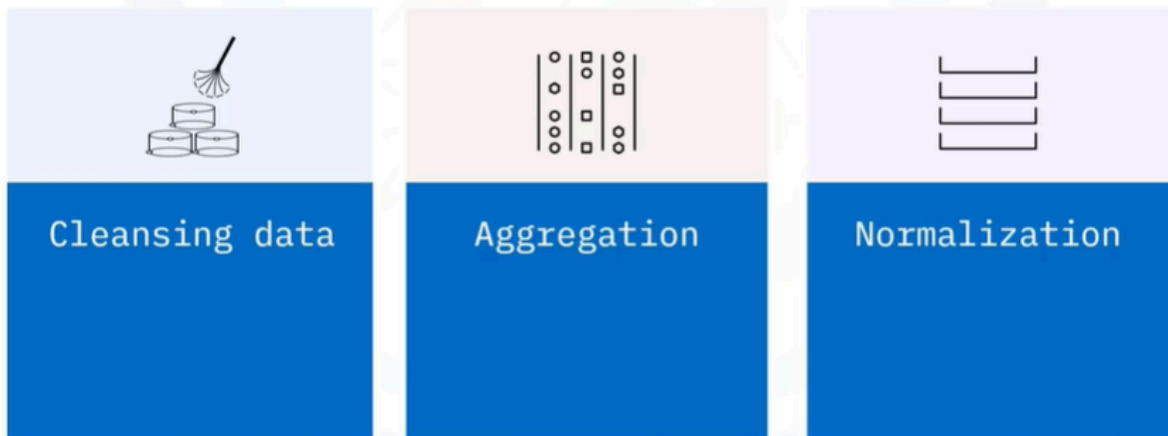
Functions of a staging area



- A staging area can have many functions. Some typical ones include:
Integration: Indeed, one of the primary functions performed by a staging area is consolidation of data from multiple source systems. Change detection: Staging areas can be set up to manage extraction of new and modified data as

needed. Scheduling: Individual tasks within an ETL workflow can be scheduled to run in a specific sequence, concurrently, and at certain times.

Functions of a staging area



- There's also: Data cleansing and validation. For example, you can handle missing values and duplicated records. Aggregating data: You can use the staging area to summarize data. For example, daily sales data can be aggregated into weekly, monthly, or annual averages, prior to loading into a reporting system. Normalizing data: To enforce consistency of data types, or names of categories such as country and state codes in place of mixed naming conventions such as "Mont," "MA," or "Montana."

Why use a staging area?

- Separate location where data from source systems is extracted to
 - Decouples operations such as validation, cleansing and other processes
 - Minimizes corruption risk
 - Simplifies ETL workflows
 - Simplifies recovery
- A staging area is a separate location, where data from source systems is extracted to. The extraction step therefore decouples operations such as validation, cleansing and other processes from the source environment. This helps to minimize any risk of corrupting source-data systems, and simplifies ETL workflow construction, operation, and maintenance. If any of the extracted data becomes corrupted somehow, you can easily recover.

Summary

In this video, you learned that staging areas:

- Are used to integrate data sources into target systems
- Can be implemented as flat files, or as tables in a database
- Decouple processing from source systems, helping to minimize data corruption
- Are often transient, also used for archiving and troubleshooting

What would happen if staging areas didn't exist in data processing?

If staging areas didn't exist in data processing, several challenges could arise:

1. **Data Corruption Risk:** Without a staging area, data would be processed directly from the source systems. This could lead to a higher risk of corrupting the original data, especially if errors occur during the transformation process.
2. **Inefficient Processing:** Data cleansing and transformation would need to happen in real-time, which could slow down the overall data processing. This could lead to delays in generating reports or insights.
3. **Complexity in ETL Workflows:** The Extract, Transform, Load (ETL) process would become more complicated. Without a staging area, it would be harder to manage and monitor the different steps of data processing, making it difficult to troubleshoot issues.
4. **Limited Data Quality Control:** A staging area allows for thorough data validation and cleansing before loading into the final data warehouse. Without it, there might be more errors, such as duplicates or missing values, in the final dataset.
5. **Difficulty in Handling Large Volumes of Data:** Staging areas can handle large amounts of data temporarily, allowing for batch processing. Without them, managing large datasets could become overwhelming and inefficient.

In summary, staging areas play a crucial role in ensuring data integrity, efficiency, and quality in the data processing workflow.

What would be the impact on data integration without staging areas?

Without staging areas, the impact on data integration would be significant:

1. **Increased Complexity:** Integrating data from multiple sources would become more complex. Without a staging area, data would need to be transformed and cleaned in real-time, making it harder to manage the integration process.
2. **Higher Error Rates:** The risk of errors would increase. Directly integrating data from source systems without a staging area could lead to issues like data corruption, duplicates, or inconsistencies, which would affect the quality of the integrated data.
3. **Slower Processing Times:** Data integration processes would likely slow down. Without a staging area to temporarily hold and process data, the system would need to handle everything in real-time, which could lead to delays in generating reports or insights.
4. **Limited Data Quality Control:** Staging areas allow for thorough data validation and cleansing before integration. Without them, it would be challenging to ensure that the data being integrated is accurate and reliable.
5. **Difficulties in Change Management:** Managing changes in data sources or structures would be more challenging. Staging areas provide a buffer that allows for adjustments and testing before final integration, which would be lost without them.

In summary, the absence of staging areas would complicate data integration, increase the likelihood of errors, and hinder the overall efficiency and quality of the data processing workflow.